MACHINE LEARNING ASSIGNMENT 1

Results Matrix — Frog Species Classification

Dataset: Frogs_MFCCs.csv | **Samples:** 7,195 | **Input features:** 22 MFCC features | **Species classes:** 10 | **Families:** 4

1. Main Results Matrix

Method	Criterion / Configuration	Train Species Accuracy	Test Species Accuracy	Family Accuracy (Test)
1. Single Decision Tree	Gini	100.00%	94.86%	96.11%
1. Single Decision Tree	Entropy	100.00%	95.34%	96.73%
2a. Bagging	Gini, 25 trees	99.97%	96.73%	97.57%
2a. Bagging	Entropy, 25 trees	99.98%	96.87%	97.50%
2b. AdaBoost SAMME	Gini, depth 3, 50 rounds	97.52%	95.34%	96.73%
2b. AdaBoost SAMME	Entropy, depth 3, 50 rounds	96.35%	94.16%	95.00%
3. Random Forest	Gini, 50 trees, max_features=4	100.00%	97.57%	97.98%
3. Random Forest	Entropy, 50 trees, max_features=4	100.00%	97.36%	97.98%
4. Hierarchical Decision Tree	Gini	100.00%	93.82%	Router: 95.07%
4. Hierarchical Decision Tree	Entropy	100.00%	94.72%	Router: 95.62%
5. Gaussian Naive Bayes	Gaussian likelihood	91.47%	90.34%	91.94%

Interpretation: The Random Forest with Gini criterion achieved the highest test species accuracy (97.57%). Family accuracy for the flat classifiers is calculated by mapping predicted species to their families. For the hierarchical model, the family column reports router accuracy, as required by the results-matrix convention.

SUPPLEMENTARY ANALYSIS

2. Hierarchical Decision Tree — Failure Analysis

Criterion	Router Accuracy	Oracle Accuracy	pA: Router Correct, Expert Wrong	pB: Router Wrong
Gini	95.07%	98.26%	1.25%	4.93%
Entropy	95.62%	98.33%	0.90%	4.38%

Definitions: Oracle accuracy measures expert classification accuracy when the correct family is supplied to the expert. pA is the fraction of test examples for which the router selects the correct family but the family expert predicts the wrong species. pB is the fraction for which the router selects the wrong family.

3. Per-Family Test Species Accuracy

Family	Test Samples	Gini Hierarchy	Entropy Hierarchy
Bufonidae	14	71.43%	71.43%
Dendrobatidae	108	88.89%	87.96%
Hylidae	433	89.84%	91.92%
Leptodactylidae	884	96.72%	97.29%

4. Hierarchical Model vs. Flat Decision Tree

Criterion	Flat Tree Test Species Accuracy	Hierarchical Test Species Accuracy	Difference (Hierarchy - Flat)
Gini	94.86%	93.82%	-1.04 percentage points
Entropy	95.34%	94.72%	-0.62 percentage points

The hierarchical model has lower overall test species accuracy than the corresponding flat decision tree under both criteria. The reported results use the shared train/test split and the outputs recorded during the assignment runs.